AnnoForge: Bootstrapping Complex Annotation from Guidelines by Recursive Self-Improvement of the Annotation Harness

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Abstract

Annotating text with a complex ontology takes both an understanding of the structures involved and close adherence to a long guideline, and at the start of a project there is too little annotated data to train a parser. We present AnnoForge, which bootstraps an annotator from the guideline and a few annotated sentences without fine-tuning. A builder LLM writes a seed harness of model steps from the guideline alone, then improves it by recursive self-improvement on groups of five gold sentences: it summarizes the errors a fine-grained judge reports, reflects on their causes, and edits the instructions, adds rule steps and writes tools over resource files such as PropBank, subject only to machine checks. On the Chinese and English test documents of the UMR parsing shared task, ten rounds (50 annotated sentences) raise a harness run by Claude Haiku from 46.4 to 52.4 (Chinese) and from 40.9 to 52.2 (English) AnCast++ F, and to 42.9 and 38.0 on the official juːmætʃ score, above the best fine-tuned shared-task system on the same documents (38.8, 27.3). The builders write PropBank lookup tools when told only that frame files exist, and the learned harness transfers unchanged to other annotators (Sonnet, GPT-6.1-sol), while the choice of generator and critic models matters far less than the harness. Editing the guideline alone fixes notation but, on Chinese, not the graphs, and reading the guideline through tools helps only a strong reasoning model. On TimeML the harness learns which events and times to link but not how they are related, and it adopts the link density of whatever gold it is trained on; an explicit linking step, useless as written by hand, becomes the best TimeML harness after one round of RSI. A guideline plus a handful of examples suffices to learn the conventions of a complex scheme and the tools it needs; judgments that no resource supports remain the model's.

1 Introduction

Annotating text with a rich ontology takes two things. One is an understanding of the structure being annotated: what an event is, how its participants and time are related, what counts as the same entity. The other is the given codebook, the guideline that fixes the labels, the conventions and the hard cases, and that annotators apply rather than invent (Halterman and Keith, 2026b; Pustejovsky and Stubbs, 2012). Complex ontologies such as Uniform Meaning Representation (UMR; Van Gysel et al., 2021; Bonn et al., 2024) and TimeML (Pustejovsky et al., 2003; Saurí et al., 2006) are hardest to annotate at the beginning, when there is little annotated data. With enough data a parser can be trained; the systems of the first UMR parsing shared task are fine-tuned models (Štěpánek et al., 2026; Heinecke and Asadullah, 2026; de Vergnette and Amblard, 2026). With a few annotated sentences that route is closed. Large language models (LLMs) offer a way to bootstrap: pretraining gives them a working understanding of meaning and of structured formats (PENMAN graphs, inline XML), and they can be given the guideline itself.

Earlier work has prepared this route. LLMs label text well when given task descriptions or codebooks (Gilardi et al., 2023; He et al., 2024; Halterman and Keith, 2026a), reasoning models can follow annotation guidelines, and an LLM can keep improving a guideline from its own errors against gold (Kim et al., 2026; Kim and Yoon, 2026; Bibal et al., 2025). These refinement studies, however, use simple and direct tasks such as named-entity recognition, whose guidelines fit in one prompt. Bootstrapping a complex scheme is different. The UMR guideline runs to 366 thousand characters, and annotating also needs outside resources such as PropBank frames (Palmer et al., 2005). A model given only the long guideline cannot follow all of it: information in the middle of a long context is used

poorly (Liu et al., 2024), accuracy falls as inputs grow (Levy et al., 2024) and as instructions accumulate (Jaroslawicz et al., 2025), and agents lose rule details of long procedure manuals (Panavas et al., 2026).

We argue that bootstrapping a complex annotation needs more than a better guideline: the guideline has to be broken into a chain of steps, and the steps need tools, so that the model's own abilities are used where they are strong and supported where they are not. We present AnnoForge (Figure 1), which builds an annotator from the guideline and a few annotated sentences by recursive self-improvement (RSI) of its *harness*: the instructions of each step (skills), the steps themselves and the tools they call. A builder model writes a seed harness from the guideline alone; then, five sentences at a time, it annotates gold sentences with the current harness, summarizes the errors, reflects on their causes and proposes edits, new steps and new tools, which are admitted by machine checks. The result annotates from instructions and a handful of examples, with no fine-tuning.

We evaluate AnnoForge on Chinese and English UMR, using the test documents of the UMR parsing shared task (Štěpánek et al., 2026), and on TimeML with the TempEval-3 platinum test set (UzZaman et al., 2013), with Claude Haiku 5.5 and GPT-6.1-sol as annotators and Claude Sonnet 5.5 as builder. Our contributions:

- A guideline-and-harness optimization method for bootstrapping complex annotation, in which skills, steps and tools are all editable and a fine-grained judge tells the builder which part of the scheme each error belongs to (§3); we also release timemljudge, a per-check judge for TimeML.
- Evidence that it works without fine-tuning: from ten groups of five annotated sentences, the learned harness improves UMR annotation on held-out shared-task documents by +6.1 (Chinese) and +11.3 (English) AnCast++ F, and exceeds the fine-tuned shared-task systems on the same documents on the official score (§5.1).
- Evidence of continued evolution: what is learned on one group carries over to unseen documents and to other annotating models; the builders invent and later revise PropBank tools when told only that frame files exist, and turn recurring convention errors into code (§5.3, §5.4). Controls show what the harness adds over a better guideline: editing the guideline alone fixes the notation but not, on Chinese, the graphs (§5.2). The analysis also shows the limit: the loop learns the conventions of the gold it sees, which pairs of events to link but not their temporal relation, and from sparse gold, to link less.

2 Related Work

LLMs as annotators, guidelines as input. LLMs can replace or assist crowd workers on many labeling tasks (Wang et al., 2021; Gilardi et al., 2023; He et al., 2024; Ziems et al., 2024; Tan et al., 2024), but they follow a scheme's conventions poorly out of the box: as AMR analyzers (Ettinger et al., 2023), as annotators of document-level event relations (Wei et al., 2024), and with real codebooks, whose underspecified concepts they silently accept (Halterman and Keith, 2026a,b); they help experts more than they replace them (Gu et al., 2026). Guidelines help when models are trained to follow them (Sainz et al., 2024; Srivastava et al., 2025) or replace examples in context (Huang et al., 2025). Guidelines can also be learned: Guideline Learning induces rules from errors (Pang et al., 2023), in the tradition of error-driven rule learning (Brill, 1995); LLMs revise annotation guidelines against gold (Bibal et al., 2025; Kim et al., 2025), and Kim et al. (2026) let a moderator model write one principle at a time into an NER guideline and keep it when development F1 improves, while Kim and Yoon (2026) refine NER instructions from the disagreements of several LLM annotators. Codebook work revises coding instructions from expert feedback or model disagreement (He et al., 2026; Yuan et al., 2026). These methods change instruction text for flat labeling tasks. Structured schemes such as UMR and TimeML are annotated by trained parsers when data exist (Chun and Xue, 2024; Markle et al., 2026; Bethard, 2013; Heinecke and Asadullah, 2026), by converting other resources (Gamba et al., 2025), or by people revising model drafts (Sun et al., 2024); we bootstrap them from the guideline and a few examples, and the object we change is the whole harness.

RSI, prompt optimization, self-improving agents and harnesses. Systems that improve the code that improves them go back to the Gödel machine (Schmidhuber, 2003); with LLMs, a model can improve its own scaffolding (Zelikman et al., 2024), an agent can rewrite its own logic (Yin et al.,

2025), and archives of self-modified agents can be searched (Zhang et al., 2026b; Novikov et al., 2025). Prompts can be searched and evolved (Zhou et al., 2023; Yang et al., 2024; Fernando et al., 2024; Guo et al., 2024), edited along textual "gradients" or execution traces (Pryzant et al., 2023; Yuksekgonul et al., 2025; Cheng et al., 2024), compiled with demonstrations (Khattab et al., 2024; Opsahl-Ong et al., 2024), evolved by reflection (Agrawal et al., 2026), or grown as playbooks (Zhang et al., 2026c; Suzgun et al., 2026); with little data they can lose to fixed prompts (Singh, 2026). Agents also learn skills (Wang et al., 2024a; Zheng et al., 2025), make tools (Cai et al., 2024; Qian et al., 2023; Yuan et al., 2024; Wang et al., 2024b), and search over workflows (Hu et al., 2025; Zhang et al., 2025; Zhuge et al., 2024). Recent work optimizes the harness of coding and web agents itself (Lee et al., 2026; Zhang et al., 2026a; Lin et al., 2026a; Xia et al., 2026), and finds that gains compound only with regression control (Wang et al., 2026) and that writing useful updates and benefiting from them dissociate across model tiers (Lin et al., 2026b). Self-feedback rarely helps without an external signal (Madaan et al., 2023; Shinn et al., 2023; Huang et al., 2024; Kamoi et al., 2024). We bring this line of work to annotation, where gold is scarce but structured: an aligned gold annotation and a judge that sorts each error into a part of the scheme give the builder far more than a pass/fail reward, and the improvement loop plays the role of the moderation step of a human annotation project (Pustejovsky and Stubbs, 2012).

3 AnnoForge

3.1 Overview

AnnoForge turns an annotation guideline and a few annotated sentences into an annotator, without training any model. Nothing in it is specific to a scheme: a benchmark supplies the guideline, the output format, a parser for that format, and an evaluator; the rest is shared (Figure 1).

Annotator. An annotator is a harness $H = (S, T, G, R)$ run by language models: *skills* S, the instructions of each model step; *tools* T, typed Python functions a step may call; a *chain* G, the order of the steps and the model that runs each; and *rule steps* R, code that rewrites an annotation without a model. A document is annotated sentence by sentence. Each step sees the document header, the five preceding sentences with their final annotations (a registry of the document so far), the sentence, the current annotation and the open findings. A *rewrite* step writes the complete annotation of the sentence, a *check* step returns findings (a problem and its fix) for the next rewrite step, a rule step is code. The final annotation is the last one that parses. Every step receives the same fixed output format; nothing else is fixed.

Seed. The builder model reads the guideline and writes the seed harness H_0: three steps, `generate` (writes the annotation), `critique` (lists problems) and `finalize` (rewrites the annotation with the findings), each a checklist of about 8,000 characters (the builder is asked for at most 8,000). It sees no annotated data and the seed has no tools.

Recursive self-improvement. Training data come as groups of five consecutive sentences of one document. Round n turns H_{n-1} into H_n with group n (§3.3): annotate the group, score it with a fine-grained judge, let the builder summarize the problems, reflect on their causes and propose operations on skills, steps and tools, admit the operations that pass machine checks, and annotate the group again. Each H_n is also tested on held-out documents, which the builder never sees.

3.2 Two benchmarks

UMR. A UMR annotation of a sentence (Van Gysel et al., 2021; Bonn et al., 2024) is a rooted graph in PENMAN notation whose nodes are concepts: events with PropBank-style senses (Palmer et al., 2005) or language-specific lemmas, entities, named-entity types, and abstract rolesets such as `have-org-role-92`; edges are participant roles (`:ARG0`, ...) and non-participant roles (`:temporal`, `:manner`, ...); attributes give aspect, polarity, mode, and the person and number of referring expressions. Each node is aligned to the words it comes from. A document-level graph per sentence adds temporal relations between events and times, modal relations between conceivers and events (who asserts what with which strength), and coreference with earlier sentences. The guideline (Van Gysel et al., 2021) has 365,742 characters.

TimeML. A TimeML annotation (Pustejovsky et al., 2003), following the TimeML 1.2.1 guidelines (Saurí et al., 2006) as in TimeBank 1.2 (Pustejovsky et al., 2006) and TempEval-3 (UzZaman et al., 2013), marks time expressions (`TIMEX3`, with

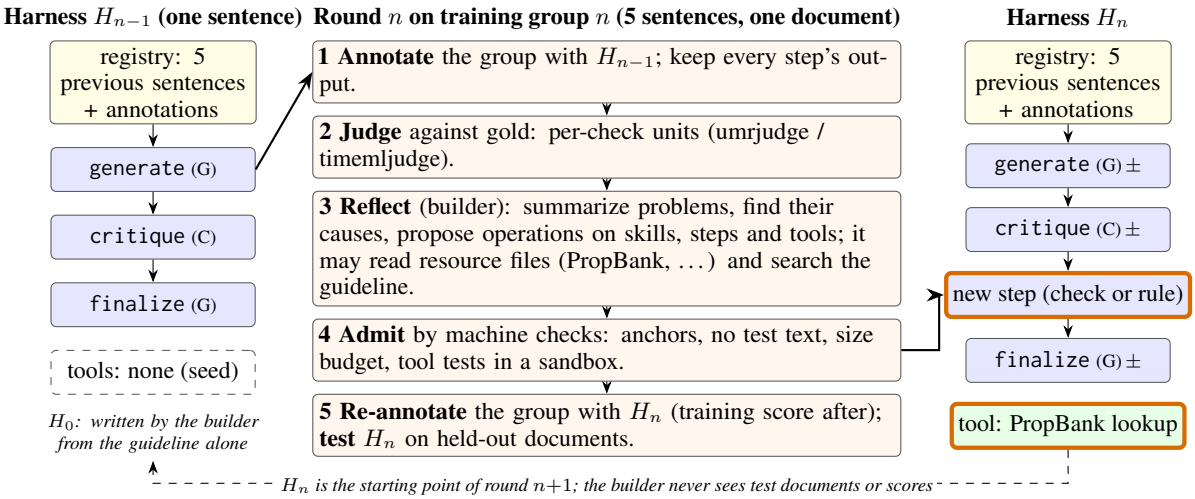


Figure 1: ANNOFORGE. Left: the seed harness annotates one sentence at a time; each step sees the previous five sentences with their annotations. Middle: one round of recursive self-improvement on a group of five gold sentences. Right: the next version; skills are edited (±), and steps and tools may be added (orange), e.g. the PropBank lookup tools the builders wrote after learning that frame files exist (§5.4). G and C are the generator and critic models of a configuration.

a type and a normalized value such as `2013-W11`), events (`EVENT`, with a class and the tense, aspect, polarity and modality of its instance), and temporal links (`TLINK`) of thirteen relation types between events, times and the document creation time, across sentences. The guideline has 115,146 characters. An annotation of a sentence is the sentence with inline tags followed by its TLINK lines.

Judges. The builder needs to know which part of the scheme an error belongs to. For UMR we use umrjudge: AnCast++ (Sun and Xue, 2024; Sun et al., 2025) aligns predicted and gold nodes, and 25 checks in 11 groups compare the aligned graphs (nodes, senses, attributes, arguments, role labels, names, reentrancy, and the document-level temporal, modal and coreference relations; Appendix B). For TimeML we built timemljudge on the same design from earlier evaluations: TempEval-3's extent and attribute measures for times and events (UzZaman et al., 2013), and the TempEval-2 split of temporal links by what they connect (Verhagen et al., 2010), event–time in one sentence or across sentences, event–document time, event–event in one sentence or across, and time–time, each scored for which pairs are linked and for the relation of the pairs linked on both sides. Both judges return *units*, each with a verdict (correct, wrong, missing, extra), the sentence, and the gold and predicted elements.

3.3 The harness and its improvement

Representation. A harness version is a directory of plain files: `harness.yaml` (the chain: each step's name, kind, model and tools), one Markdown skill per model step, one Python module per tool, one per rule step. A version is never changed after it is written; round n copies H_{n-1} and edits the copy. A configuration names two models: the generator G runs `generate` and `finalize`, the critic C runs `critique`; a step the builder adds names which of the two runs it.

Resources. The annotating steps see only their input. Tools may read resource files through one module; the builder is told what files exist (for UMR: the Chinese and English PropBank frame files, the UMR lists of abstract rolesets, relations and attribute values, and the guideline; for TimeML: the guideline) and may look into them while it reflects. It is not told what to build: in the seed there are no tools, and a PropBank lookup has to be invented.

One round. (1) *Annotate*: H_{n-1} annotates the five sentences of group n; the sentences before the group are given with their gold annotations as context. The trace keeps every step's output and findings. (2) *Judge*: the group is scored and the judge lists the units that are not correct. (3) *Reflect*: one builder call receives the output format, the harness (chain, all skills with their sizes, all tools), the resource manifest, the previous rounds' changes and training scores (never a test score), and for each sen-

tence its gold annotation, every step's output and its incorrect units. It returns *problems* (error patterns with evidence and the step to fix), a *reflection* (why they happen and what to change), *operations*, and *suggestions* it cannot implement. The builder can list and read resource files and search the guideline during the call. (4) *Admit*: operations are applied in order to a copy of H_{n-1} if they pass machine checks. (5) *Re-annotate* the group with H_n, and test H_n on held-out documents.

Operations. `edit_skill` modifies, inserts or deletes text at an anchor that must occur exactly once; `rewrite_skill` replaces a skill, to consolidate it; `add_step` inserts a check, rewrite or rule step after an existing step, `delete_step` removes one (at most five model steps); `add_tool`, `replace_tool` and `delete_tool` manage tools, and `set_step_tools` decides which steps may call which tool.

Admission. Every operation is checked by machine; there is no reviewer model, so that tools and steps are not blocked by a second opinion. No text may copy eight consecutive characters (Chinese) or six consecutive words (English) of a test sentence. A skill may not grow beyond 10,000 characters; an edit that would is rejected, and the builder is told to consolidate with `rewrite_skill`. This keeps the harness compact as rounds accumulate. A tool or rule must import only whitelisted modules (the resource module included), use no file, evaluation or reflection primitives, define one public function with type annotations and a docstring, and pass at least three tests of its own in a separate process; a tool that fails is returned once to the builder with the error. A new step must follow an existing one, and the first step must stay a rewrite step.

4 Experiments

Data. *UMR.* We use the test set of the first UMR parsing shared task (Štěpánek et al., 2026), released with UMR 2.2 (Bonn et al., 2026). Its Chinese and English documents for which we have a cleaned gold annotation [state the source and cleanup of this gold] (13 of the 15 Chinese and 4 of the 5 English test documents) are shuffled with a fixed seed and taken whole while the total stays at most 100 sentences: 7 Chinese documents (99 sentences) and the 4 English documents (95 sentences; two of them are the same text annotated twice, as in the shared task). Training groups are five consecutive sentences of one document, drawn from documents that share no sentence with the test set: Chinese UMR documents 0001–0018, and the fully annotated English documents of the release (most of it is partial AMR conversion without aspect, or Minecraft dialogue). *TimeML.* TempEval-3 (UzZaman et al., 2013): the test documents are 7 of the 20 platinum documents (96 sentences, chosen as above); training groups come from the 13 platinum documents that are not test documents, or, for a second trajectory, from TBAQ-cleaned (TimeBank 1.2 and AQUAINT, the TempEval-3 training data); all follow the TimeML 1.2.1 guidelines (Saurí et al., 2006). Each benchmark has ten training groups, used in a fixed random order, one per round; the builder never sees a test document.

Models. Annotators are Claude Haiku 5.5 (`claude-haiku-5-5`) and GPT-6.1-sol (`openai/gpt-6.1-sol` through OpenRouter); Claude Sonnet 5.5 writes the seed and is the builder in every round. Claude models run with adaptive thinking at the Messages API's default effort (Haiku medium, Sonnet high), GPT with medium reasoning effort. Each configuration was run once (answers are cached, so a repeated call is not a new sample); for UMR we repeated H_0 and H_{10} with fresh samples to estimate run-to-run variation.

Conditions. *Guideline prompt:* one call per sentence whose instructions are the whole guideline, followed by the sentence (no earlier sentences, no examples), with Haiku, Sonnet and GPT. *Seed harness H_0:* the three-step harness written from the guideline, with each of the four generator/critic pairs of Haiku and GPT, and with Sonnet in both roles. AnnoForge*:* ten rounds of RSI with Haiku as generator and critic, testing every version $H_0, \ldots, H_{10}$; the final H_{10} is also run, unchanged, with the other generator/critic pairs and with Sonnet in both roles. For TimeML we run two trajectories, with training groups from the platinum documents that are not test documents (the main one, from the same annotation project as the test set) and from TBAQ-cleaned. *Controls.* To separate what the harness adds from what a better guideline adds, we also run: the guideline *through tools*, where the instructions give only the guideline's address and the model searches and reads the guideline (and, for UMR, the resource files) with tools, with Haiku on all test documents and with GPT on two documents per benchmark; a *guideline-only round*, one

round of RSI on the first training group that starts from the guideline prompt and may only edit or rewrite it (no steps, tools or rules, no length limit); and two rounds of RSI from the seed with the tool operations disabled, and with rule steps disabled. The GPT pairs with H_{10} that the budget did not allow on all documents were run on the same two documents per benchmark. For TimeML we also test an explicit linking step (§5.4). All conditions receive the same output-format instructions.

Metrics. UMR: AnCast++ (Sun et al., 2025) sentence-graph F (*F*, the main metric) and its comprehensive score with the document-level relations (*Comp.*), after the cleanup of Chun and Xue (2024), macro-averaged over documents, and the shared task's official ju:mætʃ score (Zeman and Gamba, 2026), computed with `compare_umr.pl` against the shared task's own test gold. ju:mætʃ maps nodes through the token alignment, so it also scores alignment, and its gold differs from the cleaned gold (Appendix A). TimeML: the official TempEval-3 scorer: temporal awareness TLINK F1 (*TLINK*, the main metric), event and time-expression extent F1 and the value F1 of time expressions. The judges' per-check scores support the analysis.

Prior systems. We re-score earlier systems on exactly our test documents from their released outputs: the two UMR shared-task systems (Heinecke and Asadullah, 2026; de Vergnette and Amblard, 2026), fine-tuned on the shared task's training data, of which Orange is best on both Chinese and English; and the TempEval-3 end-to-end systems (UzZaman et al., 2013), among them ClearTK-2 (Bethard, 2013), the best participant, and TIPSem, the organizers' reference system, which is best on our documents.

Statistics. Differences between two conditions are paired over test documents (UMR: the per-document F; TimeML: the corpus-level TLINK F1 recomputed from per-document counts) with a bootstrap of 20,000 resamples; for UMR we also report a sentence-level bootstrap of per-sentence AnCast scores. With seven or four test documents and one run per configuration, the intervals describe the variation over documents, not over runs.

5 Results

5.1 Main results

Table 1 gives every system on the same test documents.

A long guideline in the prompt is not enough. Given the whole guideline and the sentence, Haiku reaches 30.9 AnCast++ F on Chinese UMR and Sonnet 50.1; on English the prompt baselines are stronger (50.8 and 54.1). Their errors follow the guideline's own notation where the data use another: modal strength written as `:modstr FullAff` inside the graph instead of document-level relations such as `(author :full-affirmative e)`, capitalized attribute values, `:ref-number` for `:refer-number`. Every prompt baseline has a document-level modal score of 0.

The seed harness helps where the guideline is hard to apply. Written from the same guideline, the three-step seed improves over the Haiku prompt baseline on Chinese UMR (46.4 vs. 30.9) and on TimeML (TLINK 39.5 vs. 30.5), where it is above the best TempEval-3 participant on our documents (ClearTK-2, 29.3) and below the organizers' TIPSem (46.3). On English UMR the seed is below the prompt baseline (40.9 vs. 50.8): a one-time compression of the guideline into 18K characters of checklists loses detail.

RSI improves the harness on held-out documents. After ten rounds of five sentences each, H_{10} improves over the seed on both UMR test sets: Chinese F 46.4 → 52.4 (paired document bootstrap +6.1, 95% interval +1.0 to +10.1; sentence level +7.3, 65 of 98 sentences better and 25 worse by more than a point) and English 40.9 → 52.2 (+11.3, +7.7 to +14.9; 72 of 95 better, 18 worse). The comprehensive score, which adds the document level, rises more (34.0 → 47.2; 22.5 → 33.0), and so does the official ju:mætʃ score against the shared task's own gold (24.1 → 42.9; 25.9 → 38.0). H_{10} is above the best shared-task system on the same documents on every UMR metric (ju:mætʃ 38.8 and 27.3; F 37.7 and 40.3), although that system was fine-tuned on the shared task's training data and we used 50 annotated sentences. Two independent repetitions of H_0 and H_{10} with fresh samples differ by 0.9/2.1 (Chinese) and 1.2/1.5 (English) F, well below the gains. On TimeML, temporal awareness does not improve (39.5 → 39.7); §5.4 shows what does change.

5.2 Guideline or harness?

Is the harness needed, or would a better guideline do? Table 2 compares, with Haiku in every role, ways of giving the model the guideline.

	UMR Chinese			UMR English			TimeML		
System	F	Comp.	ju:mæʧ	F	Comp.	ju:mæʧ	TLINK	Event	Value
Fine-tuned and feature-based systems (re-scored on our test documents)									
UMR shared task: Orange	37.7	34.0	38.8	40.3	24.8	27.3	–	–	–
UMR shared task: Sema	34.7	33.4	29.2	32.1	21.5	21.4	–	–	–
TempEval-3: TIPSem†	–	–	–	–	–	–	46.3	82.4	72.5
TempEval-3: ClearTK-2	–	–	–	–	–	–	29.3	76.9	68.0
TempEval-3: NavyTime	–	–	–	–	–	–	27.7	82.1	70.9
Guideline in the prompt, one call per sentence									
Haiku 5.5	30.9	24.1	12.5	50.8	29.9	21.7	30.5	81.5	82.6
Sonnet 5.5	50.1	38.0	24.4	54.1	31.8	28.4	35.1	83.2	80.4
GPT-6.1-sol	n/a	n/a	n/a	n/a	n/a	n/a	39.5	82.8	78.2
Seed harness H_0 (generator / critic)									
Haiku / Haiku	46.4	34.0	24.1	40.9	22.5	25.9	39.5	83.8	78.6
Haiku / GPT	46.3	33.8	25.8	43.0	24.2	26.6	38.1	81.4	70.3
GPT / Haiku	47.8	34.7	23.7	42.3	23.5	26.2	39.6	82.0	75.0
GPT / GPT	49.1	35.5	24.4	42.1	23.8	26.0	39.4	80.8	72.7
Sonnet / Sonnet	48.1	35.1	24.3	43.5	23.6	27.1	38.9	80.0	78.5
*Learned harness H_{10} (*ANNOFORGE*, built with Haiku / Haiku)*									
Haiku / Haiku	52.4	47.2	42.9	52.2	33.0	38.0	39.7	84.0	79.3
Haiku / GPT (transfer)	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
GPT / Haiku (transfer)	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
GPT / GPT (transfer)	56.4	49.7	42.4	n/a	n/a	n/a	n/a	n/a	n/a
Sonnet / Sonnet (transfer)	58.9	51.2	42.5	53.1	33.4	38.7	39.5	84.0	78.5
Haiku / Haiku, TimeBank-trained	–	–	–	–	–	–	23.8	84.2	84.1

Table 1: All systems on the test documents (UMR: 99 Chinese and 95 English sentences of the shared task's test set; TimeML: 96 TempEval-3 platinum sentences). F: AnCast++ sentence-graph F; Comp.: AnCast++ comprehensive score with document-level relations; ju:mæʧ: the shared task's official score against its test gold. TLINK: TempEval-3 temporal awareness F1; Event: event extent F1; Value: TIMEX3 value F1. Transfer: H_{10}, built with Haiku annotators, run with other generator/critic pairs. n/a: not run to completion within the API budget for GPT-6.1-sol. †TIPSem is the organizers' reference system, not a participant: the platinum gold was made by correcting TIPSem's own output (73% of the gold TLINK pairs are pairs TIPSem links, against 26% for ClearTK-2, the best participant), so its TLINK score is not comparable.

Reading the guideline through tools is worse than having it in the prompt. With only the guideline's address and tools to search and read it, Haiku falls below the prompt baseline everywhere, most on Chinese (F 14.9 vs. 30.9). It does look things up (69 %, 82 % and 24 % of its calls use a tool), but it looks up what it knows it needs, not the conventions it does not know it is missing: on Chinese it writes English concepts (`define-01`, `spaceflight`) for Chinese words. A stronger reasoning model reads better: on the two documents per benchmark of Appendix Table 8, GPT-6.1-sol with tools reaches 53.3 (Chinese) and 62.1 (English) F, as high as H_{10} with Haiku (51.4, 52.3) or with GPT (52.4, 57.1) on the same documents; the harness keeps its lead on the Chinese document level and on the official score (comprehensive 39.3 vs. 45.1, ju:mæʧ 31.0 vs. 39.2).

Editing the guideline fixes conventions, not structure. One round of reflection that may only edit the guideline prompt appends a block of "house conventions" to it (4–7K characters, at the end or right after the first line of the 366K-character guideline). This is what the harness's first round also learns, and it pays where conventions are the problem: ju:mæʧ, which scores alignment and notation, rises from 12.5 to 36.3 on Chinese and from 21.7 to 31.8 on English, and the comprehensive score rises on both. On English, where the prompt baseline is already strong, F reaches 52.5, as high as H_{10} (52.2). On Chinese, F barely moves (30.9 $\rightarrow$ 32.7) while the harness after the same round is at 47.2, and on TimeML the edited guideline (TLINK 35.5) stays below the seed harness (39.5). The difference is the decomposition: a guideline edit can say what the data's notation is, but not split the work into a draft, a check against the guideline and a revision, or give the model a lookup. Where the model's single pass is already good (English), conventions are most of what is missing; where it is not (Chinese), the steps are.

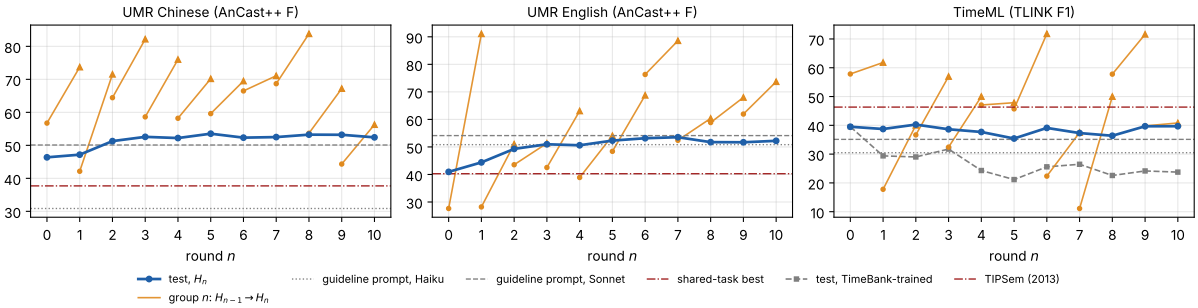


Figure 2: Ten rounds of ANNOFORGE with Haiku as generator and critic. Blue: test score of H_n. Orange: training group n, annotated by H_{n-1} (circle) and, after the round, by H_n (triangle). Grey squares (TimeML): the trajectory trained on TimeBank/AQUAINT instead of platinum documents. Reference lines: guideline-in-prompt baselines (Haiku, Sonnet) and the best earlier system on the same test documents (TimeML: TIPSem, the organizers' system whose output the platinum gold was corrected from, so not a comparable score; see Table 1).

	UMR Chinese			UMR English			TimeML		
Haiku in every role	F	Comp.	ju:mætʃ	F	Comp.	ju:mætʃ	TLINK	Event	Value
Guideline in the prompt	30.9	24.1	12.5	50.8	29.9	21.7	30.5	81.5	82.6
Guideline through tools (link only)	14.9	12.1	8.8	46.2	27.5	19.2	27.4	70.9	75.9
Guideline + 1 round (edits only)	32.7	30.0	36.3	52.5	34.3	31.8	35.5	80.5	85.4
Seed harness H_0	46.4	34.0	24.1	40.9	22.5	25.9	39.5	83.8	78.6
Harness H_1	47.2	42.4	42.0	44.4	28.4	34.3	38.7	82.4	84.7
Harness H_{10}	52.4	47.2	42.9	52.2	33.0	38.0	39.7	84.0	79.3

Table 2: The guideline against the harness. *Through tools*: the instructions give only the guideline's address; the model can search and read the guideline (and, for UMR, the resource files) with tools. *+ 1 round*: one round of RSI on the first training group in which the builder may only edit the guideline prompt (no steps, tools or rules, no length limit); H_1: the same round on the harness.

5.3 What RSI learns carries over

Each round improves its own training group, by +14.9 (Chinese), +19.1 (English) and +16.0 (TimeML) points on average (Figure 2, orange). Fitting five sentences would produce such gains on its own. For UMR they are not only that: the test score rises in most rounds (6 of 10 for Chinese, 7 for English), and across rounds the training gain and the test gain are positively correlated, though with ten rounds not significantly (Spearman ρ 0.58 and 0.50, p 0.08 and 0.14). The test curves rise for three to six groups, 15–30 annotated sentences, and then flatten (best 53.5 at round 5 for Chinese, 53.5 at round 7 for English), while the training groups keep improving: later rounds increasingly fix what is particular to a group, a narrative, a list of bond prices, a phrasal verb. Since no edit is tested before it is kept, a few rounds lower the test score, by up to two points.

Across models. H_{10} was built by watching Haiku. Run unchanged with other annotators, it helps them as much: with GPT-6.1-sol as generator and critic, Chinese F rises from 49.1 (seed) to 56.4 (comprehensive and ju:mætʃ as with Haiku, Table 1); on the 38 sentences of the two English documents the GPT run of H_{10} completed within our API budget, from 42.7 to 57.1 (Haiku: 40.5 → 52.3 on the same documents); with Sonnet in both roles, Chinese 48.1 → 58.9 and English 43.5 → 53.1. On two documents per benchmark we also ran the mixed pairs (Appendix Table 8): on UMR every pair gains 7 to 14 F from H_{10}, and the four pairs end within five points of each other; on TimeML, whose temporal-awareness score on 22 sentences moves by several points between runs, the pairs neither gain nor lose consistently. What the builder

	UMR zh (F)			UMR en (F)			TimeML (TLINK)		
G / C	H_0	H_{10}	Δ	H_0	H_{10}	Δ	H_0	H_{10}	Δ
Haiku / Haiku	46.4	52.4	+6.1	40.9	52.2	+11.3	39.5	39.7	+0.2
Haiku / GPT	46.3	n/a	n/a	43.0	n/a	n/a	38.1	n/a	n/a
GPT / Haiku	47.8	n/a	n/a	42.3	n/a	n/a	39.6	n/a	n/a
GPT / GPT	49.1	56.4	+7.3	42.1	n/a	n/a	39.4	n/a	n/a
Sonnet / Sonnet	48.1	58.9	+10.7	43.5	53.1	+9.6	38.9	39.5	+0.6

Table 3: Generator (G) and critic (C) models (H: Haiku, G: GPT-6.1-sol, S: Sonnet). H_{10} was built with Haiku in both roles; for the other pairs it is used as it is. n/a: not run within the API budget.

	UMR Chinese			UMR English		
Harness (Haiku / Haiku)	F	Comp.	ju:mætʃ	F	Comp.	ju:mætʃ
H_0 (seed)	46.4	34.0	24.1	40.9	22.5	25.9
H_{10}	52.4	47.2	42.9	52.2	33.0	38.0
H_{10} without tools	53.0	48.4	43.5	50.6	32.2	36.6
H_{10} without rule steps	54.3	49.0	43.1	52.6	32.6	37.5
H_{10} with the seed's skills	47.4	43.4	28.7	40.0	21.9	24.9

Table 4: What the parts of H_{10} contribute: the learned harness without its tools, without its rule steps, and with the seed's instructions in place of the learned ones (tools and rules kept).

wrote is about the scheme, not about the model it watched.

Generator and critic. At the seed, the choice of models matters much less than the harness (Table 3): the four Haiku/GPT pairs are within three points of each other on every benchmark, close to the run-to-run variation, while RSI adds 6 to 11. A stronger generator helps a little on Chinese (GPT/GPT 49.1 vs. Haiku/Haiku 46.4); swapping only the critic changes little (Haiku/GPT 46.3, GPT/Haiku 47.8), and the mixed pairs show no complementarity beyond that. Consistently, the builders' main complaint about the critic was not its model but its instructions: it "fixed" correct annotations (§5.5).

5.4 How the harness improves

Restricting the loop. Two rounds of RSI from the same seed with restricted operations (Appendix Table 9) agree with the ablation of H_{10}. Without rule steps the builder ends close to the full loop (Chinese 51.6 vs. 51.3 F at H_2, English 47.0 vs. 49.3), and on Chinese it writes the PropBank tool one round earlier. Without tools, Chinese falls behind (48.6), most on the official score, and English does not (49.7): in the first rounds the builder mostly fixes conventions, which instructions can carry, and the tool pays for itself on Chinese senses. With one run per configuration, differences of one or two points are within the run-to-run variation.

Conventions first, then lookups. Figure 3 breaks the gains down by judge check (Appendix Tables 13–15). For UMR, the largest gains are on conventions that the guideline states differently from the data or not at all: the document-level format and modal relations (Chinese modal $0 \rightarrow 50$, English $0 \rightarrow 32$), attribute names and values, entity concepts, granularity. The next largest are PropBank senses (Chinese $88 \rightarrow 94$, English $67 \rightarrow 80$) and core arguments of events ($52 \rightarrow 63$, $42 \rightarrow 55$). Aspect and coreference change little, and document-level temporal relations stay low (Chinese $3 \rightarrow 15$, English $5 \rightarrow 13$). The ablation (Table 4) separates the parts: the learned instructions carry most of the gain (with the seed's instructions in place of the learned ones, F falls by -5.0 and -12.3); removing the tools hardly changes F ($+0.6$, -1.6) but costs exactly the checks the tools serve, senses (Chinese $94 \rightarrow 78$, English $80 \rightarrow 73$) and core arguments ($63 \rightarrow 55$, $55 \rightarrow 47$); removing the rule steps changes little ($+1.9$, $+0.4$): the learned instructions alone already produce most of what the rules enforce. Differences of one or two points are within the run-to-run variation measured above.

TimeML: pairs, not relations. The fine-grained judge shows what the flat temporal-awareness score hides. The platinum-trained harness links the right pairs much more often: event–document time from 34 to 45, event–event within a sentence from 43 to 55, across sentences from 6 to 12 (pair F); but the relation of a pair linked on both sides gets worse (event–event within a sentence $75 \rightarrow 66$), and temporal awareness needs both. Which pairs to link is a convention of an annotation project; which relation holds is a semantic judgment. Time expressions and events, already near the 2013 systems at H_0, change little.

An explicit link step. The judge locates TimeML's problem in which pairs are linked: the seed links events to the document time that the gold leaves unlinked (123 of its 162 event–DCT links), misses most within-sentence event pairs (it links 63 of 174) and gets almost no pair across sentences right. We therefore wrote, by hand, a fourth step that rebuilds a sentence's links as a sparse backbone (a main event, one anchor per other event, the main event to the previous sentence's main event and to the document time) and a rule that links each time expression to the document time from its normalized value (Table 5). Each part on its own makes temporal awareness worse: the rule adds time–DCT links that the gold mostly does not have (only 23 of the 138 time expressions of the platinum documents are linked to the DCT, with no pattern of type or value), and the backbone links many pairs across sentences that the gold does not (36.0, and 34.0 when cross-sentence links are restricted to explicit connectives, against 39.5); added to H_{10}, the same step lowers it too (38.4, 37.3). Given to RSI, however, the step becomes something the builder

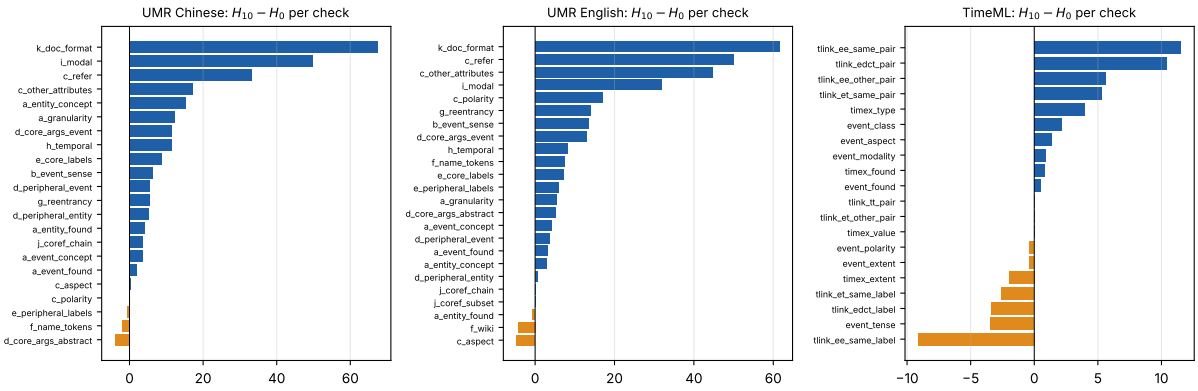


Figure 3: Change of each judge check from H_0 to H_{10} on the test documents (checks with at least six units; Haiku / Haiku; TimeML: platinum-trained trajectory).

	TLINK			E–DCT	E–E sent.	E–E across	T–DCT
Harness (Haiku)	F1	P	R	gold pairs / linked pairs (gold: 65, 174, 73, 15)			
H_0 (seed)	39.5	40.0	39.0	39/162	63/119	4/53	0/0
H_0 + DCT rule	37.3	36.4	38.2	41/162	60/107	3/55	11/46
H_0 + link step + DCT rule	36.0	28.1	50.3	20/59	60/128	32/199	8/39
H_0 + link step (revised)	34.0	30.9	37.7	25/72	62/132	3/109	0/0
+ RSI round 1	42.4	33.3	58.3	37/105	98/165	34/209	0/0
+ RSI round 2	40.4	34.6	48.4	32/97	84/147	27/160	0/0
H_{10}	39.7	42.6	37.2	32/78	84/134	8/61	0/0
H_{10} + link step + DCT rule	38.4	32.5	46.8	25/64	61/122	32/152	8/35
H_{10} + link step (revised)	37.3	37.2	37.4	26/77	65/125	2/71	0/0

Table 5: An explicit TLINK step for TimeML. The link step rebuilds the links of each sentence as a sparse backbone (main event, one anchor per event); the DCT rule links every time expression to the document time from the normalized values. Right: of the pairs the harness links, how many the gold also links.

can tune. In the first round on the platinum training group it rewrote the step after the gold's practice (dependents of a clause to its main event, the main event both to its time expression and to the document time) and reached 42.4, the best TimeML harness in our experiments (recall 58.3), above the ten-round H_{10} (39.7); in the second round it reduced the over-linking across sentences and lost part of the gain (40.4). A dedicated step does not encode the right links, but it gives the loop a place to learn them.

The builder learns the gold it sees. Trained on TimeBank and AQUAINT instead (grey in Figure 2), whose temporal links are much sparser than the platinum gold and where several five-sentence windows have no link at all, the harness learns to link less: TLINK precision rises (40.0 → 45.7), recall falls (39.0 → 16.0), and temporal awareness drops to 23.8, while time values improve (78.6 → 84.1). The loop does what it is built to do, adapt the annotator to the gold it is given; it cannot tell a project's convention from an accident of a corpus. The builders' suggestions, which the loop never applies, point the same way: 65 of the 75 suggestions of the three main trajectories are about the gold, 34 of them explicitly about inconsistencies (e.g. `:refer-number` on plural nouns in some English documents but not in others), which no learner can resolve and a human moderator should.

Tools are proposed rarely, but they are proposed. Admission was deliberately lenient: machine checks only, no reviewer. The UMR builders were told only that PropBank files exist, and each wrote a lookup in the first rounds: `lookup_frame` over the Chinese frame files (round 2), which also answers "no frame" with the UMR convention for such predicates, and `roleset_lookup` over the English argument descriptions (round 1), which the builder replaced in round 9 (and refined again in round 10) after noticing that the annotator chose wrong senses from role descriptions alone (the new version reads the XML frames with sense names and falls back from *researchers* to *research*). With H_{10}, 60 % (Chinese) and 39 % (English) of Haiku's

calls use the tool. The TimeML builder, whose only resource was the guideline, wrote no tool. Rule steps were added more often (10 for Chinese, 4 for English): the builders turned recurring convention errors into code that normalizes modal labels and constants, moves modal strength from the graph to the document level, and repairs variables and alignments. No model step was even proposed: when one kind of decision needed its own pass, the builders preferred code, or a new section in an existing skill, to an extra call per sentence. Table 6 counts every operation.

Compact by construction. Most operations modify or add instruction text; when a skill approached the 10,000-character budget, the builder consolidated it (7 rewrites for Chinese, 3 for English). The final harnesses have 26.4K (Chinese), 27.9K (English) and 28.8K (TimeML) characters of instructions, against 366K and 115K for the guidelines (Appendix Tables 10–12).

5.5 Case study: one trajectory

The English trajectory shows the order in which the builder worked (Appendix H). Round 1 found that the annotator wrote the guideline's notation (`:ref-number Singular`, modal labels such as `FullAff`, tense metanodes in temporal relations), rewrote the three skills in the data's notation, and added the PropBank tool. Rounds 2–6 worked on the document level: the `(root :modal author)` relation, modal relations for every event with aspect, coreference direction for plural pronouns, and rule steps that clean the document-level graph. Rounds 7–10 turned to semantics: aspect cues such as habitual *would*, phrasal verbs as one roleset (`pick-up-04`), the sense distinctions that led to the new lookup. The critic was a recurring problem: in several rounds the builder found that `critique` had "fixed" correct annotations and `finalize` had applied the fixes, and restricted the critic's checklist, in line with the finding that self-correction without an external signal often hurts (Huang et al., 2024; Kamoi et al., 2024).

6 Conclusion

Complex annotation can be bootstrapped from a guideline and a few examples: without fine-tuning, an LLM that is given the guideline as a chain of steps, tools over the project's resources, and a loop that revises them against a handful of gold sentences, annotates UMR better than the fine-tuned systems of its shared task on the same documents. A long guideline alone is not enough, and neither is a one-time compression of it: the gains come from the harness, its decomposition into steps and its tools, and from recursive self-improvement with a fine-grained judge, which plays the role that moderation plays in a human annotation project. In that loop the builder model learned to edit the guideline, to write the tools the guideline presupposes, and to turn recurring decisions into code.

The same loop shows its limits. What it learns are the conventions of the gold it sees: it learned which TimeML events to link, and, from a sparser corpus, to link less; senses and arguments improved where a tool supplied the frames, but judgments that no resource supports, aspect, coreference and temporal relations, moved little; and its own suggestions are dominated by inconsistencies of the gold. This points to a different division of labor in annotation projects. A guideline can be an executable object that is tested and revised against data from the first annotated sentences on; the model maintains the instructions, tools and checks, and people decide the conventions and adjudicate the inconsistencies it surfaces. Bootstrapping a new scheme or language then starts from a working annotator rather than from an empty corpus.

Limitations

Scale and variance. Each test set has at most 100 sentences in four to seven documents, and every configuration was run once, in one order of training groups, with cached answers. The bootstrap intervals describe variation over test documents, not over runs of the builder or of the annotators; repeated trajectories and other group orders are needed to estimate that variance.

Models. The builder (Sonnet) and the main annotator (Haiku) are from one model family. Within our API budget for GPT-6.1-sol (through OpenRouter, without prompt caching) we ran all seed configurations with GPT, the GPT guideline-in-prompt baseline for TimeML only (for UMR, the guideline through tools), the transfer of H_{10} to GPT/GPT for Chinese, and the other GPT pairs with H_{10} on two documents per benchmark; the cells not run on the full test sets are marked n/a. Claude models ran through the Claude Agent SDK on a subscription and GPT through OpenRouter; providers' defaults and versions may change. The controls of §5.2 and the TimeML link step were

Benchmark	edit_skill add	edit_skill modify	edit_skill delete	rewrite _skill	add _step	delete _step	add _tool	replace _tool	delete _tool	set step _tools
UMR Chinese	20/22	32/33	14/15	7/7	10/10	0	1/1	0	0	0
UMR English	24/27	32/34	18/18	3/3	4/7	0	1/1	2/2	0	0
TimeML	26/30	68/74	27/29	0	0/1	0	0	0	0	0

Table 6: Operations applied / proposed over the ten rounds, by type.

run for one or two rounds only.

Cost. An annotation with H_{10} takes about three model calls per sentence (more with tool calls). With GPT-6.1-sol in both roles it cost about $0.07 per sentence at OpenRouter's prices, so the Chinese and English part of the shared task's test set (431 sentences, of which our test sets are 194) would cost about $31 with GPT; with Haiku on a subscription it costs no money but about $0.22–0.30 per sentence at API-equivalent prices. Sentences of a document are annotated in order (each sees the five before it), so the longest document (100 sentences) bounds the wall-clock time at about two to three hours; the ten rounds of one trajectory, with a test after each, took about three hours.

Data. The English training groups come from only three fully annotated documents; most of the released English UMR data are partial conversions without aspect or document-level relations. The Chinese gold we score against differs from the shared task's own gold (Appendix A), so we report both. The TimeML results depend strongly on which gold the builder sees (§5.4). Test documents may occur in the models' pretraining data.

Admission. Operations are admitted by machine checks only; nothing tests an edit before it is kept, so a round can make the harness worse on unseen data (Figure 2). A held-in check of each edit, and people adjudicating the gold inconsistencies the builder reports, are natural additions.

Human influence. The seed builder's prompt, the reflection prompt, the operation set, the judges and the admission rules were designed by the authors. During the runs we changed the admission code twice (rule steps get one repair attempt; helper functions are allowed in tools) and restarted the trajectories from scratch each time, and we replaced the English and TimeML training pools once after finding training documents without document-level annotation.

Ethics Statement

Automating annotation can reduce the paid work available to annotators. We see ANNOFORGE as a way to start an annotation project from a working draft annotator, with people deciding conventions and adjudicating the inconsistencies the builder surfaces. UMR data are used under their CC licenses; TimeBank and AQUAINT text (LDC) is not redistributed. Model calls went to commercial APIs. Generative AI was used to run the experiments and to draft and edit this paper.

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A Data

UMR test documents. The shared task's test set (Štěpánek et al., 2026) has 15 Chinese and 5 English documents. We use those whose gold we have in a cleaned version: Chinese zh-0021–0024 and zh-0027–0035 (not zh-0026, zh-0036) and English en-0005–0008 (not en-0000). Shuffled with seed 20261012 and taken whole while the total stays at most 100 sentences, the Chinese test documents are 0023, 0027, 0028, 0030, 0031, 0032 and 0035 (99 sentences) and the English ones are en-0005, 0006, 0007 and 0008 (95 sentences). Tokens are identical to the shared task's blind test files. Our cleaned gold and the shared task's own test gold (UMR 2.2) agree at juːmætʃ 99.0 on the English documents but only 71.7 on the Chinese ones, which differ in alignment and in a number of annotation decisions; the main metric (AnCast++ F) uses the cleaned gold, juːmætʃ the official gold. The two annotations of the same English text in the test set (en-0005 and en-0007) agree at juːmætʃ 81.8 with each other, a rough ceiling for any system on that text.

UMR training groups. Chinese: ten documents of 0001–0018, one random window of five sentences each. English: the release is mostly partial AMR-to-UMR conversions (few aspect attributes, no document-level relations) or Minecraft dialogues; only three documents not sharing text with the test set have full sentence and document-level annotation (english_umr-0578 = wsj_0003, 0579 = wsj_0004, 0580 = the Pear Story), and they give the ten non-overlapping windows. The shared task's clean English training files 0001 and 0003 are the texts of test documents en-0006 and en-0008, annotated again, so we exclude them.

TimeML. Test: CNN_20130322_1003, CNN_20130322_248, CNN_20130322_314, WSJ_20130321_1145, WSJ_20130322_804, bbc_20130322_1353 and nyt_20130321_cyprus (96 sentences, scored against the official keys). Sentences are split at sentence-final punctuation followed by a capital, a digit or a quote, and at blank lines, never inside a gold tag. Training groups come either from TBAQ-cleaned (the TempEval-3 training data) or from the 13 platinum documents that are not test documents. A window is scored on its own text: its tags, and TLINKs between them or to the document creation time.

B The Judges

umrjudge. AnCast++ aligns each predicted node to at most one gold node. The judge then scores 25 checks in 11 groups over the aligned graphs; each check yields units with a verdict (correct, partial, wrong, missing, extra) and a credit, and reports F over its units.

Group	Checks
a nodes	`event_found`, `event_concept`, `entity_found`, `entity_concept`, `granularity`
b senses	`event_sense`
c attributes	`aspect`, `modstr`, `polarity`, `refer`, `other_attributes`
d arguments	`core_args_event`, `core_args_abstract`, `peripheral_event`, `peripheral_entity`
e role labels	`core_labels`, `peripheral_labels`
f names	`name_tokens`, `wiki`
g reentrancy	`reentrancy`
h–k document	`temporal`, `modal`, `coref_chain`, `coref_subset`, `doc_format`

Table 7: The checks of umrjudge.

timemljudge. Predicted and gold `TIMEX3` and `EVENT` tags are matched one-to-one by character overlap, largest first. Times: `timex_found` (relaxed extent), `timex_extent` (exact extent of a matched pair), `timex_type`, `timex_value`. Events: `event_found`, `event_extent`, `event_class`, and the instance attributes `event_tense`, `event_aspect`, `event_polarity`,

`event_modality`. Temporal links, with predicted ends mapped to their matched gold entities and B–A links turned around: for event–time in the same sentence and in different sentences, event–document time, event–event in the same and in different sentences, and time–time, a `pair` check (are the right pairs linked) and a `label` check (the relation of a pair linked on both sides). The first group follows TempEval-3's entity measures (UzZaman et al., 2013), the link categories TempEval-2's tasks C–F (Verhagen et al., 2010); unlike the official temporal-awareness score (UzZaman and Allen, 2011), the link checks use no temporal closure.

C Prompts

Output format (UMR). Every annotating step receives this system message (the TimeML one is analogous): *"You annotate {Chinese|English} text in Uniform Meaning Representation (UMR), one sentence at a time. Output format. The annotation of sentence number N is written exactly as a .umr file writes one sentence, in three sections:* `# sentence level graph:` *(a PENMAN graph; every variable is sN followed by a letter and optional digits)* `# alignment:` *(one line per variable:* `variable: start-end`*, 1-based word positions;* `0-0` *when aligned to no word)* `# document level annotation: (sNs0 / sentence :temporal (. . .) :modal (. . .) :coref (. . .))`. *In the document level annotation, a and b are variables of this sentence or of earlier sentences, or constants; leave out a group that has no relation."*

Seed. The builder receives the description of the three steps, the output format and the whole guideline, and is asked for each step's instructions, "at most 8,000 characters, in English, organized as a compact checklist a model can follow: what to annotate and in which order, the decisions that matter most for this scheme and how the guidelines settle them, the closed lists of labels and values that apply, and the frequent mistakes to avoid".

Reflection. The reflection prompt (`af/rsi.py` in the code) describes the annotator, the resource files and the operations with their constraints, then gives the current harness, the history of earlier rounds (training scores, applied and rejected operations), and the group's sentences with gold, every step's output and the judge's incorrect units. It ends: "problems: the error patterns of this group, most costly first, each with the step it should be fixed in. reflection: why they happen, and what you will change and why. operations: the changes, in the order to apply them. suggestions: anything else." Suggestions are never implemented, and the prompt says so.

	UMR zh (F)		UMR en (F)		TimeML (TLINK)	
	H_0	H_{10}	H_0	H_{10}	H_0	H_{10}
Guideline in the prompt, Haiku	26.5		58.3		29.8	
Guideline through tools, Haiku	19.9		50.9		27.1	
Guideline through tools, GPT	53.3		62.1		36.0	
Harness, Haiku / Haiku	43.1	51.4	40.5	52.3	37.8	36.9
Harness, Haiku / GPT	38.4	52.8	43.1	53.7	31.3	29.0
Harness, GPT / Haiku	42.4	53.2	42.4	55.3	34.3	38.1
Harness, GPT / GPT	45.0	52.4	42.7	57.1	32.8	n/a

Table 8: Two test documents per benchmark (22, 38 and 22 sentences), on which every generator / critic pair (Haiku, GPT-6.1-sol) was run with the seed and with H_{10} (built with Haiku), and the guideline baselines. TimeML is scored by the official scorer on the two documents.

D Reproduction

All code, harness versions, per-round reflections, operations, diffs and scores are in the experiment folder `experiment/annoforge` of the code repository [anonymize for review]; `af/run.py` annotates and scores one harness version, `af/rsi.py` runs a trajectory, `af/sota.py` re-scores earlier systems, and `analysis/annoforge.py` in the paper repository makes every table and figure. Claude models ran through the Claude Agent SDK, GPT-6.1-sol through OpenRouter; every answer is cached, so a run can be replayed exactly. TimeBank and AQUAINT text is not redistributed.

E Additional Results

Table 8 compares every generator/critic pair on two test documents per benchmark, with the guideline baselines on the same documents; Table 9 gives the two-round RSI runs with restricted operations (§5.4).

F Round-by-Round Records

Tables 10–12 list every round of the three main trajectories: the main metric of the training group before and after the round (F for UMR, TLINK F1 for TimeML), the test score of the new version, the operations applied out of those proposed, the total size of the skills, and the number of model steps, rule steps and tools.

		UMR Chinese			UMR English			tools/rules	
RSI with		F	Comp.	juːmæʧ	F	Comp.	juːmæʧ	zh	en
all operations	H_1	47.2	42.4	42.0	44.4	28.4	34.3	0/1	1/0
	H_2	51.3	46.6	42.2	49.3	30.7	36.8	1/1	1/0
no tools	H_1	45.5	41.4	33.0	42.9	26.8	33.8	0/0	0/0
	H_2	48.6	44.0	36.9	49.7	32.6	35.6	0/0	0/0
no rule steps	H_1	50.8	45.8	36.3	43.3	27.4	33.2	1/0	1/0
	H_2	51.6	48.8	45.1	47.0	28.7	36.4	2/0	1/0

Table 9: RSI with restricted operations (two rounds, Haiku / Haiku, from the same seed and training groups). tools/rules: tools and rule steps admitted so far.

	Group n		Test	Ops	Skills	Model steps /
	H_{n-1}	H_n	H_n	applied	(K chars)	rules / tools
H_0	–	–	46.4	–	17.4	3 / 0 / 0
H_1	56.7	73.7	47.2	4/4	13.0	3 / 1 / 0
H_2	42.1	71.5	51.3	4/4	12.0	3 / 1 / 1
H_3	64.4	82.2	52.6	10/10	15.6	3 / 2 / 1
H_4	58.6	76.0	52.2	5/5	17.8	3 / 3 / 1
H_5	58.2	70.2	53.5	5/6	18.9	3 / 4 / 1
H_6	59.6	69.5	52.3	18/18	19.6	3 / 5 / 1
H_7	66.5	71.1	52.5	9/11	20.1	3 / 6 / 1
H_8	68.7	83.8	53.2	9/9	21.7	3 / 7 / 1
H_9	53.7	67.2	53.2	12/13	23.6	3 / 8 / 1
H_{10}	44.3	56.3	52.4	8/8	26.4	3 / 10 / 1

Table 10: Rounds of ANNOFORGE on Chinese UMR (AnCast++ F).

G Per-Check Scores

H Case Study Excerpts

The builder's reflection, English round 1 (verbatim, abridged).

> "Most of the lost score is format, not understanding. The skills taught a vocabulary (`:ref-person`, `Singular`, `:modstr`, `:modpred`, `:Str`, `DCT`) that the gold and the judge do not use. The critique step then reinforced it, and it even moved the temporal relation the wrong way. I will change four things. (1) I rewrite all three skills to use the gold conventions [...] (2) I add a deterministic rule step [...] that normalises attribute names and cases, converts `:modstr` into the document-level modal labels and then drops it [...] (3) I add a `roleset_lookup` tool over `propbank-en-args.txt`, so haiku picks real senses and roles instead of guessing [...] (4) I rewrite the critic to check against the same conventions."

The rule steps of (2) were rejected: after one repair attempt they still came with fewer than three tests. The builder added rule steps that passed in later rounds.

A tool and its revision (English). Round 1, the whole tool:

```
def roleset_lookup(lemma: str) -> str:
  """Look up English PropBank rolesets for a lemma
[. . . ]. Returns lines like 'dump-01 ARG0: ...
```

	Group n		Test	Ops	Skills	Model steps /
	H_{n-1}	H_n	H_n	applied	(K chars)	rules / tools
H_0	–	–	40.9	–	17.9	3 / 0 / 0
H_1	27.6	91.1	44.4	4/6	14.4	3 / 0 / 1
H_2	28.2	51.0	49.3	14/15	17.0	3 / 0 / 1
H_3	43.5	51.3	51.0	11/11	18.4	3 / 1 / 1
H_4	42.5	63.1	50.6	4/5	19.3	3 / 1 / 1
H_5	38.9	54.1	52.3	6/6	21.8	3 / 2 / 1
H_6	48.4	68.8	53.2	10/11	24.8	3 / 3 / 1
H_7	76.3	88.6	53.5	7/7	26.3	3 / 3 / 1
H_8	52.4	60.4	51.7	7/9	26.0	3 / 3 / 1
H_9	58.9	68.0	51.7	7/8	26.9	3 / 3 / 1
H_{10}	61.9	73.7	52.2	14/14	27.9	3 / 4 / 1

Table 11: Rounds of ANNOFORGE on English UMR (AnCast++ F).

	Group n		Test	Ops	Skills	Model steps /
	H_{n-1}	H_n	H_n	applied	(K chars)	rules / tools
H_0	–	–	39.5	–	17.2	3 / 0 / 0
H_1	57.8	61.8	38.7	13/15	19.9	3 / 0 / 0
H_2	17.8	40.0	40.3	11/11	21.3	3 / 0 / 0
H_3	36.7	57.0	38.6	17/17	22.5	3 / 0 / 0
H_4	32.4	50.0	37.7	13/15	23.6	3 / 0 / 0
H_5	47.1	47.9	35.4	10/11	24.2	3 / 0 / 0
H_6	45.7	71.8	39.1	13/13	24.7	3 / 0 / 0
H_7	22.4	37.5	37.3	14/15	25.7	3 / 0 / 0
H_8	11.1	50.0	36.4	10/11	27.3	3 / 0 / 0
H_9	57.8	71.6	39.7	11/16	28.3	3 / 0 / 0
H_{10}	39.8	40.8	39.7	9/10	28.8	3 / 0 / 0

Table 12: Rounds of ANNOFORGE on TimeML, trained on platinum documents (TLINK F1).

```
ARG1: ...' with all senses and their numbered
roles [. . . ]."""
  lem = lemma.strip().lower().replace(' ',
'-').replace('_', '-')
  text =
resources.read_resource('propbank-en-args.txt')
  out = [ln for ln in text.splitlines() if
ln.startswith(lem + '-')]
  return '\n'.join(out[:14]) if out else 'no
roleset for ' + lem
```

Round 9, reason given: "Show sense names (meaning) and roles from the XML, with fallback to base verb for derived nouns/adjectives, so the model can choose the right sense (consider-01 vs -02)." The new version reads `propbank-en/<lemma>.xml`, prints each roleset as `consider-02 [name] ARG0: . . .`, and tries *-ers*, *-ion*, *-ing* and other suffix variants when the lemma has no file.

A rule step (Chinese, round 10, the whole function).

```
def run(annotation: str, sentence: dict) -> str:
  """Ensure the document-level :modal group
contains (root :modal author), creating the group
if absent."""
  [. . . ] if PAT.search(doc): return annotation
  m = re.search(r':modal\s*\(', doc)
  if m: doc = doc[:m.end()] + '(root :modal
author)\n' + doc[m.end():]
  else: [add a :modal group with it] . . .
```

Group	Check	Units	H_0	H_{10}	Δ
a	a_event_found	471	94.4	96.6	+2.2
a	a_event_concept	421	81.2	84.9	+3.6
a	a_entity_found	1278	89.5	93.7	+4.2
a	a_entity_concept	1036	65.9	81.2	+15.3
a	a_granularity	1308	83.6	96.0	+12.3
b	b_event_sense	322	87.6	93.8	+6.2
c	c_aspect	392	74.5	74.8	+0.4
c	c_polarity	26	76.2	76.2	+0.0
c	c_refer	11	0.0	33.3	+33.3
c	c_other_attributes	280	52.6	69.9	+17.2
d	d_core_args_event	333	51.8	63.5	+11.6
d	d_core_args_abstract	50	39.2	35.2	−4.0
d	d_peripheral_event	191	35.0	40.6	+5.6
d	d_peripheral_entity	354	43.3	48.5	+5.2
e	e_core_labels	374	78.3	87.1	+8.8
e	e_peripheral_labels	309	77.3	76.7	−0.7
f	f_name_tokens	191	58.1	56.2	−1.9
f	f_wiki	97	0.0	–	–
g	g_reentrancy	340	30.0	35.5	+5.5
h	h_temporal	431	3.0	14.7	+11.6
i	i_modal	507	0.0	49.8	+49.8
j	j_coref_chain	275	13.8	17.5	+3.6
j	j_coref_subset	4	0.0	0.0	+0.0
k	k_doc_format	1437	23.0	90.5	+67.5

Table 13: umrjudge checks on the Chinese test documents, H_0 and H_{10} (Haiku / Haiku).

A Chinese tool answering with a convention. `lookup_frame(lemma)` returns the rolesets of a Chinese PropBank frame file, and when there is none: "NO FRAME for *lemma*: use *lemma*-00 with `:actor` (agent) / `:undergoer` (patient) instead of ARGn", the UMR guideline's convention for predicates without a frame file, attached by the builder to the tool's answer.

Group	Check	Units	H_0	H_{10}	Δ
a	a_event_found	546	90.5	93.6	+3.1
a	a_event_concept	451	76.1	80.1	+4.1
a	a_entity_found	817	89.0	88.4	−0.6
a	a_entity_concept	655	77.4	80.2	+2.8
a	a_granularity	1046	88.4	93.9	+5.5
b	b_event_sense	315	66.7	80.1	+13.4
c	c_aspect	365	64.3	59.6	−4.7
c	c_polarity	31	45.0	61.9	+16.9
c	c_refer	114	0.0	50.0	+50.0
c	c_other_attributes	495	11.4	56.1	+44.7
d	d_core_args_event	341	41.6	54.5	+13.0
d	d_core_args_abstract	55	38.7	43.8	+5.1
d	d_peripheral_event	150	25.2	28.8	+3.7
d	d_peripheral_entity	174	42.7	43.4	+0.6
e	e_core_labels	366	76.8	84.0	+7.2
e	e_peripheral_labels	162	64.2	70.1	+5.9
f	f_name_tokens	143	71.7	79.2	+7.5
f	f_wiki	127	71.0	66.7	−4.3
g	g_reentrancy	354	18.9	33.0	+14.1
h	h_temporal	418	4.8	13.1	+8.3
i	i_modal	520	0.0	32.0	+32.0
j	j_coref_chain	427	17.2	17.4	+0.2
j	j_coref_subset	62	0.0	0.0	+0.0
k	k_doc_format	1239	24.7	86.2	+61.5

Table 14: umrjudge checks on the English test documents.

Group	Check	Units	H_0	H_{10}	Δ
a timex	timex_found	62	89.3	90.1	+0.8
a timex	timex_extent	50	92.0	90.0	−2.0
a timex	timex_type	50	94.0	98.0	+4.0
a timex	timex_value	50	88.0	88.0	+0.0
b event	event_found	384	83.8	84.3	+0.5
b event	event_extent	277	100.0	99.6	−0.4
b event	event_class	277	83.0	85.2	+2.2
b event	event_tense	277	68.2	64.8	−3.5
b event	event_aspect	277	88.8	90.2	+1.3
b event	event_polarity	277	100.0	99.6	−0.4
b event	event_modality	277	95.3	96.2	+0.9
c tlink E-T	tlink_et_same_pair	66	74.3	79.6	+5.3
c tlink E-T	tlink_et_same_label	39	89.7	87.2	−2.6
c tlink E-T	tlink_et_other_pair	15	0.0	0.0	+0.0
d tlink E-DCT	tlink_edct_pair	188	34.4	44.8	+10.4
d tlink E-DCT	tlink_edct_label	39	84.6	81.2	−3.4
e tlink E-E	tlink_ee_same_pair	230	43.0	54.5	+11.5
e tlink E-E	tlink_ee_same_label	63	74.6	65.5	−9.1
e tlink E-E	tlink_ee_other_pair	122	6.3	11.9	+5.6
e tlink E-E	tlink_ee_other_label	4	75.0	50.0	−25.0
f tlink T-T	tlink_tt_pair	15	0.0	0.0	+0.0

Table 15: timemljudge checks on the TimeML test documents (platinum-trained trajectory).